CAMBRIDGE INTERNATIONAL EXAMINATIONS

International General Certificate of Secondary Education

MARK SCHEME for the October/November 2012 series

0620 CHEMISTRY

0620/22 Paper 2 (Core Theory), maximum raw mark 80

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1 **(a)** **(i)** D / phosphorus / P; [1]

(ii) E / helium / He; [1]

(iii) C / chlorine / Cl_2 / Cl; [1]

(iv) A / copper / Cu; [1]

(v) A / copper / Cu; [1]

(b) C; D; [2]

(c) giant; covalent; [2]

(d) substance containing only 1 type of atom / substance which cannot be broken down into a simpler one; [1]
allow: substance which can't be separated by chemical means
ignore: substance with one atom / substance with similar types of atom

[Total: 10]

2 **(a)** (damp) red litmus (paper); [1]
turns blue; [1]
note: second mark dependent on correct reagent
allow: universal indicator (1 mark); turns blue / purple (1 mark)
allow: 1 mark for litmus paper turns blue / pH paper turns blue
allow: white fumes (1 mark); with hydrochloric acid vapour (1 mark)
ignore: other chemicals added as long as it is clear that ammonia is the gas being tested

(b) pH 9; [1]

(c) **(i)** NH_4Cl on right; [1]

(ii) structure completely correct;; [2]
allow: 1 mark for 1 pair of electrons bonded between H and Cl
ignore: inner shell electrons

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(d) (i) any 4 of: [4]
use of burette
add indicator to flask
add acid to alkali (or vice versa)
until indicator changes colour
record volume (of acid or alkali added) **ignore:** amount of acid or alkali added
repeat without indicator
using same volume of acid and ammonia as in previous experiment

(ii) heat to crystallisation (point) / evaporate some of the water / leave to crystallise [1]
allow: heat then cool
ignore: heat (unqualified) / heat to dryness / heat to get rid of all the water

[Total: 11]

3 (a) (i) get darker / deeper colour; [1]

(ii) gas; [1]
allow: answer written in table

(iii) any value between –180 to –20 °C (actual = –101 °C); [1]
allow: answer written in table

(b) (i) chlorine → bromine → iodine → astatine;; [2]
allow: 1 mark if one pair incorrect way round / order completely reversed

(ii) no **and** chlorine is more reactive (than bromine) / bromine is less reactive; [1]
ignore: chlorine is very reactive / bromine is not very reactive
ignore: chloride is more reactive

(c) H_2O (on right); [1]
2 on left (this is dependent on H_2O being the product); [1]

(d) (i) to kill bacteria / to kill microbes / to disinfect it [1]
allow: to kill germs / to get rid of bacteria
ignore: to clean water

(ii) any two of: [2]
minerals or (dead) remains insoluble in water
these particles are large / water particles (molecules) are very small
(larger particles) get stuck (between the sand particles) / (larger particles) remain in the sand / trapped by sand
water (molecules) drain through / water comes out the bottom
ignore: water is filtered

[Total: 11]

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4 **(a)** groups of hydrocarbons / molecules; [1]
with similar (range of) boiling points / sizes / masses; [1]
allow: 1 mark for idea of separating molecules for particular fuels
ignore: petroleum broken down / smaller molecules formed / mixture of fuels

(b) **(i)** gasoline; diesel; [2]

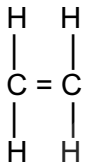
(ii) refinery gas: heating / cooking; [1]
allow: fuel
bitumen: roads / roofing; [1]

(c) high temperature; [1]
allow: heat / stated temperature of 200 °C or more
catalyst; [1]
ignore: name of catalyst
ignore: pressure

(d) **(i)** substance containing hydrogen and carbon only; [1]

(ii) C_4H_8 / $2C_2H_4$; [1]

(e) **(i)** [1]



(ii) monomers; addition; polymers; [3]

[Total: 14]

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5 **(a)** any two of; [2]
Al has low density / iron has high density
allow: lightweight or light for density)
Al does not form coloured compounds / iron formed coloured compounds
Al has only one oxidation state / iron has several oxidation states
Al does not act as a catalyst / iron can act as a catalyst
Al is softer / iron is harder (comparative needed)
Al has lower density / iron has higher density (comparative needed)
Al is a better conductor / iron is not as good a conductor (comparative needed)
Al is weaker / iron is stronger (comparative needed)
ignore: melting and boiling points

(b) any suitable use e.g. aircraft or car (bodies) / food containers / pots and pans / electrical wiring / drinks cans; [1]

(c) precipitate formed; [1]
which is white in colour; [1]
dissolves (in excess sodium hydroxide); [1]
allow: precipitate disappears

[Total: 6]

6 **(a)** **(i)** limestone / chalk; [1]

(ii) the other product is a gas / carbon dioxide escapes; [1]
allow: carbon dioxide is a gas / waste gases are gone / CO_2 formed
allow: reaction goes completely to the right

(b) **(i)** $C + O_2 \rightarrow CO_2$;; [2]
allow: 1 mark for O_2 as reactant / $C + 2O \rightarrow CO_2$

(ii) limited; air; monoxide; poisonous; [4]
allow: oxygen in place of air
note: if dioxide put in third position allow 1 mark for harmless in 4th position

(c) calcium chloride; [1]
water; [1]

(d) **(i)** idea of measure the (decrease in) mass / weight; [1]
idea of measuring time (intervals); [1]

(ii) increases / faster; [1]
decreases / slower; [1]
increases / faster; [1]
note: the answers above must be comparative
allow: 1 mark for fast; slow; fast
ignore: reference to time taken

[Total: 15]

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7 **(a)** **(i)** any 4 of: [4]
(at 20 °C / at the start) particles are close together / touching / arranged regularly
(at 20 °C / at the start) particles are vibrating / not moving
as temperature rises / then particles vibrate more / gain energy
at 114 °C / then particles begin to move
forces between particles weaken / molecules start to break away (from each other)
at 114 °C / then particles become more randomly arranged / slide over each other
when liquid / above 114 °C / then particles slide over each other/ move
when liquid / above 114 °C then particles are randomly arranged
ignore: particles further apart / particles (move) faster

(ii) 254; [1]

(b) **(i)** ionic; [1]

(ii) KI; [1]

(c) (1 each) [4]
insoluble / does not dissolve; doesn't conduct;
soluble / dissolves; doesn't conduct;
ignore: low / high / not very well

(d) + electrode: iodine / I_2 / I; [1]
– electrode: potassium / K; [1]
allow: 1 mark if correct electrode products reversed
ignore: iodide

[Total: 13]

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